

FRANCESCO GRUSSU

CONTACT AND ONLINE PROFILES

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EDUCATION AND TRAINING

PhD in MR Physics	2012-2016	University College London (UCL), UK Mark: viva passed with no corrections. Award date: 28/03/2016. Thesis: " <i>Microstructural imaging of the human spinal cord with advanced diffusion MRI</i> "
Master's Degree in Bioengineering	2010-2012	University of Genoa, Italy Mark: 110 out of 110 <i>cum laude</i> & <i>Dignità di Stampa</i> (Examination Panel award). Dissertation: " <i>A study on a bidirectional brain-machine interface inspired by the corticospinal control of movement</i> ".
Bachelor's Degree in Biomedical Engineering	2006-2009	University of Cagliari, Italy Mark: 110 out of 110 <i>cum laude</i> . Dissertation: " <i>Real time wavelet denoising on a DSP of neural signals coming from the peripheral nervous system</i> ".

WORK EXPERIENCE

Vall d'Hebron Institute of Oncology	Jan.2025-now	Senior investigator, VHIO — SPAIN Affiliations: Radiomics Group, Biomedical Research Department. Role: supervisor of research line in diffusion MRI. Design of MRI studies.
Vall d'Hebron Institute of Oncology	Oct.2020-Dec.24	Senior post-doc, VHIO — BARCELONA, SPAIN Affiliations: Radiomics Group, Clinical Research Department. Role: development of microstructural MRI techniques in cancer.
University College London	Sep.2020-now	Honorary Senior Fellow, UCL — LONDON, UK Affiliations: Queen Square Institute of Neurology. Role: collaborator in MRI development in multiple sclerosis.
University College London	Feb.2016-Sept.20	Research Associate, UCL — LONDON, UK Affiliations: Institute of Neurology; Centre for Medical Image Computing. Responsibilities: quantitative MRI development (spinal cord, brain, prostate).
	May-June 2012	Research Assistant, QMUL — LONDON, UK

Queen Mary Univ.
of London

Affiliation: School of Electronic Engineering and Computer Science (EECS).
Responsibilities: image segmentation (Computed Tomography Angiography).

RESEARCH VISITS

Visiting Researcher	Feb.-March 2024	Champalimaud Foundation, Portugal
	Affiliation: Preclinical MRI lab, Champalimaud Foundation, Lisbon, Portugal. Training: preclinical MRI and multi-echo gradient echo imaging.	
Visiting Researcher	Sept.-Nov. 2017	New York University (NYU), USA
	Affiliation: Radiology, Langone Medical Center, New York City (USA). Training: advanced denoising techniques for diffusion MRI.	

IMPACT STATS ON THE 27/07/2026

<i>h-index</i>	Google Scholar h-index: 26. Web of Science h-index: 21.
<i>Citations</i>	Total Google Scholar citations: 2807. Web of Science citations: 1658.

FUNDING: PROJECT GRANTS

EU4Health 2026	EU4Health project grant , call EU4H-2026-SANTE-PJ-o8, European Commission. <i>"Scale-up biotech AI innovation by leveraging responsible data use under the European Health Data Space"</i> (SCAILED). Award: 92,362.4 €. Role: co-investigator for VHIO. Principal investigator and primary coordinator: Gu Wei, Luxembourg National Data Service (LNDS). Consortium: 47 institutes/centres, with total budget of 14,384,759.99 €. Duration: 01/10/2026-30/09/2029.
CaixaResearch 2025	CaixaResearch Institute Innovation Hub 2025 . CaixaResearch Institute, Barcelona (Spain). <i>"Computational Magnetic Resonance for metastatic prostate cancer"</i> . Award: 30,000 €. Role: principal investigator. Duration: 01/01/2026-31/12/2026. Competitive call open to researchers from Innovation Hub institutions (VHIO; IrsiCaixa; ISGlobal; BarcelonaBeta).
AEI Generación de Conocimiento/ Knowledge Generation 2024	2024 Knowledge Generation Projects (Proyectos de Generación de Conocimiento) , National Research Agency (Agencia Estatal de Investigación (AEI)), Ministry of Science, Innovation and Universities, Spain. <i>"Magnetic Resonance Imaging (MRI) foundational artificial intelligence for non-invasive, histologically-meaningful cancer characterisation (MRI-Found-Histo)"</i> . Award: 118,750 €. Role: principal investigator. Duration: 01/09/2025-31/08/2028. Code: PID2024-158670OA-Ioo.
UCL pump-priming award 2017	Departmental seed grant . Centre for Medical Image Computing <i>Pump-priming Award</i> at University College London (UCL). <i>"Enabling multi-site high precision spinal cord MRI"</i> . Award: GBP 23,900. Role: principal investigator. Duration: 01/07/2017-30/06/2018.

FUNDING: FELLOWSHIPS AND STUDENTSHIPS

ISCIII "Miguel Servet" Fellowship 2026	2026 Miguel Servet tenure-track fellowship , Carlos III Health Institute (ISCIII, Instituto de Salud Carlos III, Ministry of Health and Ministry of Science, Innovation and Universities, Spain). <i>"Bridging histology and medical imaging with biophysical digital twins to enhance precision oncology"</i> . Award: €308,600. Role: principal investigator and fellow. Duration: 2027-2031 (provisional award 15/07/2026). Code: CP26/00050.
AEI "Ramón y Cajal" Fellowship 2025	2025 Ramón y Cajal tenure-track fellowship , National Research Agency (Agencia Estatal de Investigación (AEI)), Ministry of Science, Innovation and Universities,

Spain. “Computational Magnetic Resonance biophysics for enhanced neurological and oncological imaging”. **Award:** €259,810. **Role:** principal investigator and fellow. **Duration:** 2025-2030 (awarded 10/06/2026). **Code:** RYC2025-054665-I.

“la Caixa” Junior
Leader Fellowship
2022

2022 Junior Leader Retaining post-doctoral fellowship, “la Caixa” Foundation, Spain. “New-generation oncological MRI (New-OncoMRI): development, validation and application”. **Award:** €294,900. **Role:** principal investigator and fellow. **Duration:** 30/09/2022-29/09/2025. **Code:** ID 100010434, fellowship number LCF/BQ/PR22/11920010.

AGAUR Beatriu
de Pinós
Fellowship 2020

2020 Beatriu de Pinós post-doctoral fellowship, AGAUR, Secretary of Universities and Research (Govt of Catalonia, Spain). “Advancing Magnetic Resonance Imaging against liver cancer”. **Award:** €144,300. **Role:** principal investigator and fellow. **Duration:** 01/01/2022-31/12/2024, renounced on 29/09/2022 due to incompatibility with the “la Caixa” Junior Leader fellowship. **Code:** 2020 BP 00117.

UCL Grand
Challenge PhD
studentship 2012

UCL School of Life and Medical Science *Grand Challenge PhD Studentship*, awarded to work on the project “Axonal density as MR imaging biomarker: from bench to bedside”. **Award:** 59,000 GBP. **Description:** PhD studentship covering a 3-year student stipend, tuition fees at UK/EU rate, and research costs. **Duration:** 09/2012-12/2015.

POSITIONS: DECLINED COMPETITIVE AWARDS

2025 UAB “Serra
Húnter”
tenure-track
Lecturership

Serra Húnter tenure-track Lecturership / Assistant Professorship (*professor ajudant doctor*) funded by the Generalitat de Catalunya. Department of Computer Sciences, Universitat Autònoma de Barcelona, Cerdanyola/Barcelona, Spain. **Duration:** 2026-2032 (awarded 30/06/2026, declined due to incompatibility with RyC fellowship). **Code:** UAB-LE-251-010.

SCIENTIFIC ARTICLES AND REVIEWS — SENIOR AUTHORSHIP

Magn Reson Med
2026

“Simulation-informed evaluation of microvascular parameter mapping for diffusion MR imaging of solid tumours”. Voronova AK et al, Grussu F. *Magnetic Resonance in Medicine* (2026), 96(1): 387-402, doi: [10.1002/mrm.70318](https://doi.org/10.1002/mrm.70318). Senior (last) author, corresponding author. **IF:** 3.3 (Quartile Q2) in the year of publication.

Comm Bio 2025

“Histology-informed microstructural diffusion simulations for MRI cancer characterisation — the Histo-microSim framework”. Grigoriou A et al, Grussu F. *Communications Biology* (2025), 8: 1695, doi: [10.1038/s42003-025-09096-3](https://doi.org/10.1038/s42003-025-09096-3). Senior (last) author, corresponding author. **IF:** 5.1 (Decile D1) in the year of publication.

MedIA 2025

“SpinFlowSim: a blood flow simulation framework for histology-informed diffusion MRI microvasculature mapping in cancer”. Voronova AK et al, Grussu F. *Medical Image Analysis* (2025), 102: 103531, doi: [10.1016/j.media.2025.103531](https://doi.org/10.1016/j.media.2025.103531). Senior (last) author, corresponding author. **IF:** 11.8 (Decile D1) in the year of publication.

JMRI 2025

“Enhancing tumor microstructural quantification with machine learning and diffusion-relaxation MRI”. Macarro C et al, Grussu F. *Journal of Magnetic Resonance Imaging* (2025), 61(2): 1018-1021, doi: [10.1002/jmri.29484](https://doi.org/10.1002/jmri.29484). Senior (last) author, corresponding author. **IF:** 3.5 (Quartile Q1) in the year of publication.

JMRI 2024

Review: “Advanced diffusion-weighted MRI for cancer microstructure assessment in body imaging, and its relationship with histology”. Fokkinga E, Hernandez-Tamames JA, Ianus A, Nilsson M, Tax CMW, Perez-Lopez R, Grussu F. *Journal of Magnetic Resonance Imaging* (2024), 60(4): 1278-1304, doi: [10.1002/jmri.29144](https://doi.org/10.1002/jmri.29144). Senior (last) author, corresponding author. **IF:** 3.3 (Quartile Q1) in the year of publication.

Front Neurol 2021

"Comparison of neurite orientation dispersion and density imaging and two-compartment spherical mean technique parameter maps in multiple sclerosis". Johnson D et al, Grussu F. *Frontiers in Neurology* (2021), 12: 662855, doi: [10.3389/fneur.2021.662855](https://doi.org/10.3389/fneur.2021.662855). Senior (last) author. IF: 4.086 (Quartile Q2) in the year of publication.

SCIENTIFIC ARTICLES AND REVIEWS — (CO)-FIRST AUTHORSHIP

J Immunother
Cancer 2025

"Integrating C-reactive protein flare and early MRI dynamics for enhanced prediction of immunotherapy response". Navarro D*, Grussu F* et al. *Journal for ImmunoTherapy of Cancer* (2025), 13: e012143, doi: [10.1136/jitc-2025-012143](https://doi.org/10.1136/jitc-2025-012143). *: DN and FG are joint first authors (equal contribution). IF: 10.6 (Decile D1) in the year of publication.

Comm Med 2025

"Clinically feasible liver tumour cell size measurement through histology-informed in vivo diffusion MRI". Grussu F et al. *Communications Medicine* (2025), 5: 535, doi: [10.1038/s43856-025-01246-2](https://doi.org/10.1038/s43856-025-01246-2). Corresponding author. IF: 6.3 (Quartile Q1) in the year of publication.

Cell Rep Med 2024

"An accessible deep learning tool for voxel-wise classification of brain malignancies from perfusion MRI". Garcia-Ruiz A*, Pons-Escoda A*, Grussu F* et al. *Cell Reports Medicine* (2024), 5(3): 101464, doi: [10.1016/j.xcrm.2024.101464](https://doi.org/10.1016/j.xcrm.2024.101464). *: AGR, APE and FG are joint first authors (equal contribution). IF: 11.7 (Decile D1) in the year of publication.

Magn Reson Med
2022

"Diffusion MRI signal cumulants and hepatocyte microstructure at fixed diffusion time: Insights from simulations, 9.4T imaging, and histology". Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(1): 365-379, doi: [10.1002/mrm.29174](https://doi.org/10.1002/mrm.29174). Corresponding author. IF: 3.737 (Quartile Q2) in the year of publication.

Front Phys 2021

"Feasibility of data-driven, model-free quantitative MRI protocol design: application to brain and prostate diffusion-relaxation imaging". Grussu F et al. *Frontiers in Physics* (2021), 9: 752208, doi: [10.3389/fphy.2021.752208](https://doi.org/10.3389/fphy.2021.752208). Corresponding author. IF: 3.560 (Quartile Q2) in the year of publication.

Sem Ultrasound
CT MRI 2021

Review: *"Diffusion-weighted imaging: recent advances and applications".* Martinez-Heras E*, Grussu F*, et al. *Seminars in Ultrasound, CT and MRI* (2021), 42(5): 490-506, doi: [10.1053/j.sult.2021.07.006](https://doi.org/10.1053/j.sult.2021.07.006). *: EMH and FG are joint first authors (equal contribution). IF: 1.600 (Quartile Q3) in the year of publication.

NeuroImage 2020

"Multi-parametric quantitative in vivo spinal cord MRI with unified signal readout and image denoising". Grussu F et al. *NeuroImage* (2020), 217: 116884, doi: [10.1016/j.neuroimage.2020.116884](https://doi.org/10.1016/j.neuroimage.2020.116884). Corresponding author. IF: 5.902 (Decile D1) in the year of publication.

Magn Reson Med
2019

"Relevance of time-dependence for clinically viable diffusion imaging of the spinal cord". Grussu F et al. *Magnetic Resonance in Medicine* (2019), 81(2): 1247-1264, doi: [10.1002/mrm.27463](https://doi.org/10.1002/mrm.27463). Corresponding author. IF: 3.858 (Quartile Q1) in the year of publication.

Ann Clin Transl
Neurol 2017

"Neurite dispersion: a new marker of multiple sclerosis spinal cord pathology?". Grussu F, Schneider T et al. *Annals of Clinical and Translational Neurology* (2017), 4(9):663-679, doi: [10.1002/acn3.445](https://doi.org/10.1002/acn3.445). FG and TS are joint first authors (equal contribution). IF: 3.901 (Quartile Q1) in the year of publication. Paper featured in *Nature Reviews Neurology "Research Highlights"* (Patel M, *Nat Rev Neur* (2017), 13(10): 578, doi: [10.1038/nrneurol.2017.127](https://doi.org/10.1038/nrneurol.2017.127)).

J Neurosci Meth
2016

"A framework for optimal whole-sample histological quantification of neurite orientation dispersion in the human spinal cord". Grussu F et al. *Journal of Neuroscience Methods* (2016), 273: 20-32, doi: [10.1016/j.jneumeth.2016.08.002](https://doi.org/10.1016/j.jneumeth.2016.08.002). Corresponding author. IF: 2.053 (Quartile Q3) in the year of publication.

NeuroImage 2015

"Neurite orientation dispersion and density imaging of the healthy cervical spinal cord

in vivo". Grussu F et al. *NeuroImage* (2015), 111: 590-601, doi: [10.1016/j.neuroimage.2015.01.045](https://doi.org/10.1016/j.neuroimage.2015.01.045). Corresponding author. IF: 6.357 (Decile D1) in the year of publication.

SCIENTIFIC ARTICLES AND REVIEWS — CO-AUTHORSHIP

- Eur Radiol* 2026 **Review:** "ESR Essentials: diffusion-weighted MRI—practice recommendations by the European Society for Magnetic Resonance in Medicine and Biology". Palombo M, Bodini B, Grussu F et al. *European Radiology* (2026), 36: 2198-2208, doi: [10.1007/s00330-025-12033-x](https://doi.org/10.1007/s00330-025-12033-x). IF: 4.7 (Quartile Q1) in the year of publication.
- JMRI* 2026 "The sense of smell (SoS) atlas: its creation and first application to investigate COVID-19 related anosmia with a comprehensive quantitative MRI protocol". Gaviraghi M, ..., Grussu F et al. *Journal of Magnetic Resonance Imaging* (2026), 63(2): 574-593, doi: [10.1002/jmri.70128](https://doi.org/10.1002/jmri.70128). IF: 3.5 (Quartile Q1) in the year of publication.
- Clin Cancer Res* 2026 "Broad utility of ultrasensitive analysis of ctDNA dynamics across solid tumors treated with immunotherapy". Garralda E, ..., Grussu F et al. *Clinical Cancer Research* (2026), 32(2): 333-349, doi: [10.1158/1078-0432.CCR-25-2312](https://doi.org/10.1158/1078-0432.CCR-25-2312). IF: 10.2 (Decile D1) in the year of publication.
- Magn Res Med* 2025c **Review:** "Considerations and recommendations from the ISMRM Diffusion Study Group for preclinical diffusion MRI: Part 3—Ex vivo imaging: Data processing, comparisons with microscopy, and tractography". Schilling KG, Howard AFD, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2561-2582, doi: [10.1002/mrm.30424](https://doi.org/10.1002/mrm.30424). IF: 3.0 (Quartile Q2) in the year of publication.
- Magn Res Med* 2025b **Review:** "Considerations and recommendations from the ISMRM diffusion study group for preclinical diffusion MRI: Part 2—Ex vivo imaging: Added value and acquisition". Schilling KG, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2535-2560, doi: [10.1002/mrm.30435](https://doi.org/10.1002/mrm.30435). IF: 3.0 (Quartile Q2) in the year of publication.
- Magn Res Med* 2025a **Review:** "Considerations and recommendations from the ISMRM diffusion study group for preclinical diffusion MRI: Part 1: In vivo small-animal imaging". Jelescu IO, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2507-2534, doi: [10.1002/mrm.30429](https://doi.org/10.1002/mrm.30429). IF: 3.0 (Quartile Q2) in the year of publication.
- Imag Neurosci* 2025 "Body size and intracranial volume interact with the structure of the central nervous system: A multi-center in vivo neuroimaging study". Labounek R, ... Grussu F et al. *Imaging Neuroscience* (2025), 3: 00559, doi: [10.1162/imag.a.00559](https://doi.org/10.1162/imag.a.00559). No IF available in the year of publication.
- Sci Rep* 2025 "Evaluation of magnetic resonance spectroscopy total sodium concentration measures, and associations with microstructure and physical impairment in cervical myelopathy". Solanky B, ... Grussu F et al. *Scientific Reports* (2025), 15: 7014, doi: [10.1038/s41598-025-91658-w](https://doi.org/10.1038/s41598-025-91658-w). IF: 3.9 (Quartile Q1) in the year of publication.
- J Immunother Cancer* 2025 "Radiomics signature for dynamic monitoring of tumor inflamed microenvironment and immunotherapy response prediction". Bernatowicz K, ... Grussu F et al. *Journal for ImmunoTherapy of Cancer* (2025), 13: e009140, doi: [10.1136/jitc-2024-009140](https://doi.org/10.1136/jitc-2024-009140). IF: 10.6 (decile D1) in the year of publication.
- NeuroImage Rep* 2024 "Investigating the relationship between thalamic iron concentration and disease severity in secondary progressive multiple sclerosis using quantitative susceptibility mapping: Cross-sectional analysis from the MS-STAT2 randomised controlled trial". Williams T, ..., Grussu F, ..., Chataway J; On behalf of The UCL MS-STAT2 investigators. *NeuroImage: Reports* (2024), 4(3): 100216, doi: [10.1016/j.ynirp.2024.100216](https://doi.org/10.1016/j.ynirp.2024.100216). No IF available in the year of publication.
- Eur Urol* 2024 "Whole-body magnetic resonance imaging as a treatment response biomarker in castration-resistant prostate cancer with bone metastases: the iPROMET clinical trial".

Garcia-Ruiz A, ... Grussu F et al. *European Urology* (2024), 86(3): 272-274, doi: [10.1016/j.eururo.2024.02.016](https://doi.org/10.1016/j.eururo.2024.02.016). IF: 25.3 (Decile D1) in the year of publication.

- Mult Scler* 2024 "What contributes to disability in progressive MS? A brain and cervical cord-matched quantitative MRI study". Tur C, ..., Grussu F et al. *Multiple Sclerosis Journal* (2024), e-pub ahead of print, doi: [10.1177/13524585241229969](https://doi.org/10.1177/13524585241229969). IF: 4.8 (Quartile Q1) in the year of publication.
- Radiology: AI* 2024 "Identification of precise 3D CT radiomics for habitat computation by machine learning in cancer". Prior O, ..., Grussu F et al. *Radiology: Artificial Intelligence* (2024), 6(2): e230118, doi: [10.1148/ryai.230118](https://doi.org/10.1148/ryai.230118). IF: 8.1 (Quartile Q1) in the year of publication.
- Movement Disord* 2023 "Multimodal analysis of the visual pathways in Friedreich's Ataxia reveals novel biomarkers". Thomas-Black G, ... Grussu F et al. *Movement Disorders* (2023), 38(6): 959-969, doi: [10.1002/mds.29277](https://doi.org/10.1002/mds.29277). IF: 8.6 (Decile D1) in the year of publication.
- Sci Rep* 2023 "Feasibility of in vivo multi-parametric quantitative magnetic resonance imaging of the healthy sciatic nerve with a unified signal readout protocol". Boonsuth R, Battiston M, Grussu F et al. *Scientific Reports* (2023), 13: 6565, doi: [10.1038/s41598-023-33618-w](https://doi.org/10.1038/s41598-023-33618-w). IF: 4.6 (Quartile Q2) in the year of publication.
- Front Neuroinform* 2023 "Patterns of inflammation, microstructural alterations, and sodium accumulation define multiple sclerosis subtypes after 15 years from onset". Ricciardi A, Grussu F et al. *Frontiers in Neuroinformatics* (2023), 17: 1060511, doi: [10.3389/fninf.2023.1060511](https://doi.org/10.3389/fninf.2023.1060511). IF: 3.5 (Quartile Q2) in the year of publication.
- Neurology* 2023 "Differentiating Multiple Sclerosis from AQP4-Neuromyelitis Optica Spectrum Disorder and MOG-antibody disease with imaging". Cortese R, ... Grussu F et al. *Neurology* (2023), 100(3): e308-e323, doi: [10.1212/WNL.0000000000001465](https://doi.org/10.1212/WNL.0000000000001465). IF: 10.1 (Decile D1) in the year of publication.
- Am J Neuroradiol* 2022 "Diffuse large B-cell Epstein-Barr virus-positive primary CNS lymphoma in non-AIDS patients: high diagnostic accuracy of DSC perfusion metrics". Pons-Escoda A, ... Grussu F et al. *American Journal of Neuroradiology* (2022), 43(11): 1567-1574, doi: [10.3174/ajnr.A7668](https://doi.org/10.3174/ajnr.A7668). IF: 4.966 (Quartile Q2) in the year of publication.
- Magn Reson Med* 2022c "Multi-echo quantitative susceptibility mapping: how to combine echoes for accuracy and precision at 3 Tesla". Biondetti E, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(5): 2101-2116, doi: [10.1002/mrm.29365](https://doi.org/10.1002/mrm.29365). IF: 3.737 (Quartile Q2) in the year of publication.
- Magn Reson Med* 2022b "SENSE EPI reconstruction with 2D phase error correction and channel-wise noise removal". Powell E, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(5): 2157-2166, doi: [10.1002/mrm.29349](https://doi.org/10.1002/mrm.29349). IF: 3.737 (Quartile Q2) in the year of publication.
- Magn Reson Med* 2022a "Comparison of multicenter MRI protocols for visualizing the spinal cord gray matter". Cohen-Adad J, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(2): 849-859, doi: [10.1002/mrm.29249](https://doi.org/10.1002/mrm.29249). IF: 3.737 (Quartile Q2) in the year of publication.
- Eur Radiol* 2022 "Voxel-level analysis of normalized DSC-PWI time-intensity curves: a potential generalizable approach and its proof of concept in discriminating glioblastoma and metastasis". Pons-Escoda A, ..., Grussu F et al. *European Radiology* (2022), 32: 3705-3715, doi: [10.1007/s00330-021-08498-1](https://doi.org/10.1007/s00330-021-08498-1). IF: 7.034 (Quartile Q1) in the year of publication.
- BMJ Open* 2022 "Histo-MRI map study protocol: a prospective cohort study mapping MRI to histology for biomarker validation and prediction of prostate cancer". Singh S, ..., Grussu F et al. *BMJ Open* (2022), 12: e059847, doi: [10.1136/bmjopen-2021-059847](https://doi.org/10.1136/bmjopen-2021-059847). IF: 3.007 (Quartile Q2) in the year of publication.
- NeuroImage Clin* 2022 "Spatial patterns of brain lesions assessed through covariance estimations of lesional

- voxels in multiple sclerosis: the SPACE-MS technique*". Tur C, Grussu F et al. *NeuroImage: Clinical* (2022), 33: 102904, doi: [10.1016/j.nicl.2021.102904](https://doi.org/10.1016/j.nicl.2021.102904). IF: 4.891 (Quartile Q2) in the year of publication.
- Front Neurol* 2021 "Assessing lumbar plexus and sciatic nerve damage in relapsing-remitting multiple sclerosis using magnetisation transfer ratio". Boonsuth R, ..., Grussu F et al. *Frontiers in Neurology* (2021), 12: 763143, doi: [10.3389/fneur.2021.763143](https://doi.org/10.3389/fneur.2021.763143). IF: 4.003 (Quartile Q2) in the year of publication.
- Sci Rep* 2021 "Robust imaging habitat computation using voxel-wise radiomics features". Bernatowicz K, Grussu F et al. *Scientific Reports* (2021), 11: 20133, doi: [10.1038/s41598-021-99701-2](https://doi.org/10.1038/s41598-021-99701-2). IF: 4.380 (Quartile Q1) in the year of publication.
- Sci Data* 2021 "Open-access quantitative MRI data of the spinal cord and reproducibility across participants, sites and manufacturers". Cohen-Adad J, ..., Grussu F et al. *Scientific Data* (2021), 8: 219, doi: [10.1038/s41597-021-00941-8](https://doi.org/10.1038/s41597-021-00941-8). IF: 6.444 (Quartile Q1) in the year of publication.
- Nat Protoc* 2021 "Generic acquisition protocol for quantitative MRI of the spinal cord". Cohen-Adad J, ..., Grussu F et al. *Nature Protocols* (2021), 16: 4611–4632, doi: [10.1038/s41596-021-00588-0](https://doi.org/10.1038/s41596-021-00588-0). IF: 13.491 (decile D1) in the year of publication.
- Brain* 2021 "Brain microstructural and metabolic alterations detected in vivo at the onset of the first demyelinating event". Collorone S, ..., Grussu F et al. *Brain* (2021), 144: 1409–1421, doi: [10.1093/brain/awab043](https://doi.org/10.1093/brain/awab043). IF: 13.501 (Decile D1) in the year of publication.
- NeuroImage* 2021 "Uncertainty modelling in deep learning for safer neuroimage enhancement: demonstration in diffusion MRI". Tanno R, ... Grussu F et al. *NeuroImage* (2021), 225: 117366, doi: [10.1016/j.neuroimage.2020.117366](https://doi.org/10.1016/j.neuroimage.2020.117366). IF: 6.556 (Decile D1) in the year of publication.
- Mult Scler* 2020b "Reduced neurite density in the brain and cervical spinal cord in relapsing–remitting multiple sclerosis: A NODDI study". Collorone S, Cowley N, Grussu F et al. *Multiple Sclerosis Journal* (2020), 26(13): 1647–1657, doi: [10.1177/1352458519885107](https://doi.org/10.1177/1352458519885107). IF: 5.412 (Quartile Q1) in the year of publication.
- Mult Scler* 2020a "A multi-shell multi-tissue diffusion study of brain connectivity in early multiple sclerosis". Tur C, Grussu F et al. *Multiple Sclerosis Journal* (2020), 26(7): 774–785, doi: [10.1177/1352458519845105](https://doi.org/10.1177/1352458519845105). IF: 5.412 (Quartile Q1) in the year of publication.
- NeuroImage* 2020b "Cross-scanner and cross-protocol multi-shell diffusion MRI data harmonization: Algorithms and results". Ning L, Bonet-Carne E, Grussu F et al. *NeuroImage* (2020), 221: 117128, doi: [10.1016/j.neuroimage.2020.117128](https://doi.org/10.1016/j.neuroimage.2020.117128). IF: 5.902 (Decile D1) in the year of publication.
- NeuroImage* 2020a "Generalised boundary shift integral for longitudinal assessment of spinal cord atrophy". Prados F, ..., Grussu F et al. *NeuroImage* (2020), 209: 116489, doi: [10.1016/j.neuroimage.2019.116489](https://doi.org/10.1016/j.neuroimage.2019.116489). IF: 5.902 (Decile D1) in the year of publication.
- Magn Reson Med* 2019 "Fast bound pool fraction mapping via steady-state magnetization transfer saturation using single-shot EPI". Battiston M, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2019), 82: 1025–1040, doi: [10.1002/mrm.27792](https://doi.org/10.1002/mrm.27792). IF: 3.858 (Quartile Q1) in the year of publication.
- NeuroImage* 2019 "Cross-scanner and cross-protocol diffusion MRI data harmonisation: a benchmark database and evaluation of algorithms". Tax CMW, Grussu F et al. *NeuroImage* (2019), 195: 285–299, doi: [10.1016/j.neuroimage.2019.01.077](https://doi.org/10.1016/j.neuroimage.2019.01.077). IF: 5.812 (Decile D1) in the year of publication.
- Sci Rep* 2018 "Structural cortical network reorganization associated with early conversion to multiple sclerosis". Tur C, ..., Grussu F et al. *Scientific Reports* (2018), 8: 10715, doi: [10.1038/s41598-018-29017-1](https://doi.org/10.1038/s41598-018-29017-1). IF: 4.122 (Quartile Q1) in the year of publication.

- Magn Reson Med* 2018b “An optimized framework for quantitative magnetization transfer imaging of the cervical spinal cord *in vivo*”. Battiston M, Grussu F et al. *Magnetic Resonance in Medicine* (2018) 79(5): 2576-2588, doi: [10.1002/mrm.26909](https://doi.org/10.1002/mrm.26909). IF: 3.924 (Quartile Q1) in the year of publication.
- Magn Reson Med* 2018a “Fast and reproducible *in vivo* T1 mapping of the human cervical spinal cord”. Battiston M, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2018), 79(4): 2142-2148, doi: [10.1002/mrm.26852](https://doi.org/10.1002/mrm.26852). IF: 3.924 (Quartile Q1) in the year of publication.
- NeuroImage* 2017 “Spinal cord grey matter segmentation challenge”. Prados F, ..., Grussu F et al. *NeuroImage* (2017), 152:312-329, doi: [10.1016/j.neuroimage.2017.03.010](https://doi.org/10.1016/j.neuroimage.2017.03.010). IF: 5.835 (Decile D1) in the year of publication.
- PlosOne* 2016 “Reduced field-of-view diffusion-weighted imaging of the lumbosacral enlargement: a pilot *in vivo* study of the healthy spinal cord at 3T”. Yiannakas MC, Grussu F et al. *PlosOne* (2016), 11(10): e0164890, doi: [10.1371/journal.pone.0164890](https://doi.org/10.1371/journal.pone.0164890). IF: 3.057 (Quartile Q1) in the year of publication.

CONFERENCE PAPERS: FIRST AUTHORSHIP

- CDMRI* 2020 “Deep learning model fitting for diffusion-relaxometry: a comparative study”. Grussu F, Battiston M, Palombo M, Schneider T, Wheeler-Kingshott CAM, Alexander DC. *Proc of 2020 MICCAI Workshop on Computational Diffusion MRI*, 2021, 159-172, doi: [10.1007/978-3-030-73018-5_13](https://doi.org/10.1007/978-3-030-73018-5_13). Corresponding author.

CONFERENCE PAPERS: CO-AUTHORSHIP

- MICCAI* 2022 “Progressive subsampling for oversampled data - application to quantitative MRI”. Blumberg SB, ..., Grussu F et al. *Proc of Medical Image Computing and Computing Assisted Intervention (MICCAI) 2022, Lecture Notes in Computer Science*, 13436: 421-431, doi: [10.1007/978-3-031-16446-0_40](https://doi.org/10.1007/978-3-031-16446-0_40).
- CDMRI* 2019 “Acquiring and predicting multidimensional diffusion (MUDI) data: an open challenge”. Pizzolato M, ..., Grussu F et al. *Proc of 2019 MICCAI Workshop on Computational Diffusion MRI*, 2020, 195-208, doi: [10.1007/978-3-030-52893-5_17](https://doi.org/10.1007/978-3-030-52893-5_17).
- CDMRI* 2018b “Multi-shell diffusion MRI harmonisation and enhancement challenge (MUSHAC): progress and results”. Ning L, ..., Grussu F et al. *Proc of 2018 MICCAI Workshop on Computational Diffusion MRI*, 2019, 217-224, doi: [10.1007/978-3-030-05831-9_18](https://doi.org/10.1007/978-3-030-05831-9_18).
- CDMRI* 2018a “Spatial characterisation of fibre response functions for spherical deconvolution in multiple sclerosis”. Tur C, Grussu F et al. *Proc of 2018 MICCAI Workshop on Computational Diffusion MRI*, 2019, 265-279, doi: [10.1007/978-3-030-05831-9_21](https://doi.org/10.1007/978-3-030-05831-9_21).
- MICCAI* 2016 “Bayesian image quality transfer”. Tanno R, ..., Grussu F et al. *Proc of Medical Image Computing and Computing Assisted Intervention (MICCAI) 2016, Lecture Notes in Computer Science*, 9901: 265-273, doi: [10.1007/978-3-319-46723-8_31](https://doi.org/10.1007/978-3-319-46723-8_31).

BOOK CHAPTERS

- CRC Press* 2018 Chapter 8: “D — the diffusion of water (DTI)”. Grussu F and Wheeler-Kingshott CAM. “Quantitative MRI of the brain” (2nd edition, 2018), Cercignani M, Dowell N and Tofts P editors. ISBN 978-1-138-03285-9, doi: [10.1201/b21837](https://doi.org/10.1201/b21837).

BOOK EDITING

- CDMRI 2018 “Computational Diffusion MRI”. Bonet-Carne E, Grussu F, Ning L, Sepehrband F and Tax C editors. Proc. of 2018 MICCAI Workshop on “Computational Diffusion MRI”, Granada, Spain, 20/09/2018. ISBN: 978-3-030-05830-2, doi: [10.1007/978-3-030-05831-9](https://doi.org/10.1007/978-3-030-05831-9).
- CDMRI 2017 “Computational Diffusion MRI”. Kaden E, Grussu F, Ning L, Tax C and Veraart J editors. Proc. of 2017 MICCAI Workshop on “Computational Diffusion MRI”, Quebec City, Canada, 10/09/2017. ISBN: 978-3-319-73839-0, doi: [10.1007/978-3-319-73839-0](https://doi.org/10.1007/978-3-319-73839-0).

PRIZES AND AWARDS

- 2023 **Best oral paper award**, 2023 annual meeting of the **Iberian Chapter** of the International Society for Magnetic Resonance in Medicine (ISMRM), Trainee competition for abstract Grussu F et al, Proc of ISMRM Iberian Chapt. 2023.
- 2021 **3rd prize**, 2021 ISMRM **MR of Cancer Study Group**, Trainee competition for abstract Grussu F et al, Proc of ISMRM 2021, p.0699.
- 2021 *Magna cum Laude* award, abstract (Grussu et al, p.0699, ISMRM 2021).
- 2020 *Magna cum Laude* award, abstract (Grussu et al, p.1035, ISMRM 2020).
- 2020 **2nd prize** (shared), 2020 ISMRM British and Irish Chapter “*Mansfield Research Innovation Award*” for abstract Grussu F et al, Proc of ISMRM 2020.
- 2019 **1st prize** in the “Multi-dimensional Diffusion Imaging” (**MUDI**) challenge at 2019 CDMRI MICCAI Workshop (Shenzhen, China, 17/10/2019) (**Team**: Grussu F, Blumberg SB, Ianus A, Mertzanidou T, Alexander DC; **Method**: SARDU-Net).
- 2018-2020 **Elected trainee representative** for the *White Matter Study Group* of the International Society for Magnetic Resonance in Medicine (ISMRM).
- 2018 & 2019 “**Distinguished reviewer**” Award for Magnetic Resonance in Medicine, awarded at the 2018 and 2019 ISMRM annual meetings.
- 2018 *Magna cum Laude* award, abstract (Grussu et al, p.466, ISMRM 2018).
- 2017 Poster short-listed for presentation at the 2017 ISMRM *Diffusion study group* (Grussu et al, p.3399, ISMRM 2017).
- 2017 *Magna cum Laude* award, abstract (Grussu et al, p.3399, ISMRM 2017).
- 2016 Abstract submission among best 5 in the “*Validation*” session, ISMRM workshop *Breaking the barriers of diffusion MRI*.
- 2016 Poster short-listed for presentation at the 2016 ISMRM *Diffusion study group* (Grussu et al, p.2009, ISMRM 2016).
- 2015 Young Investigators poster competition finalist (80 selected), European Committee for Treatment and Research in Multiple Sclerosis (ECTRIMS) 2015.
- 2015 *Magna cum Laude* award, abstract (Grussu et al, p.0909, ISMRM 2015).
- 2015 *Magna cum Laude* award, abstract (Grussu et al, p.0154, ISMRM 2015).
- 2013 **1st prize**, Master’s degree awards, Mòguru council, Italy (1st prize).
- 2013 Poster short-listed as a finalist of the 2013 ISMRM *White Matter Study Group* poster competition.
- 2012 **2012 Master’s degree thesis prize**, *Gruppo Nazionale di Bioingegneria* (GNB, National Bioengineering Group).
- 2009 **Award for the best student graduating in Biomedical Engineering** (BEng) in

2009, University of Cagliari, Italy.

2008-2011 **"Assegno di Merito" (Merit cheque)** by the Sardinian regional government for excellence in Academic Studies, obtained yearly from 2008 to 2011.

AWARDS IN SUPERVISORY ROLES

- 2024 **Best application (poster)** by the Diffusion Study Group of the International Society for Magnetic Resonance in Medicine (ISMRM) to student Anna Voronova, for abstract (Voronova A et al, and Grussu F, p.0124, ISMRM 2024).
- 2024 **Summa cum Laude** award to student Anna Voronova, for abstract (Voronova A et al, and Grussu F, p.0124, ISMRM 2024).
- 2024 **Magna cum Laude** award to student Athanasios Grigoriou, for abstract (Grigoriou A et al, and Grussu F, p.0699, ISMRM 2024).

TRAVEL GRANTS

- 2020 UCL Dept. of neuroinflammation travel grant funding attendance at the 2020 ISMRM meeting (GBP 400) – awarded for April 2020, unused due to COVID-19.
- 2016 ISMRM Trainee Stipend for attendance at ISMRM workshop *"Breaking the barriers of Diffusion MRI (waived registration fees)"*.
- 2015, 2016 UCL School of Life and Medical Sciences Travel Grant funding the attendance at the 2015 (GBP 800) and 2016 (GBP 940) ISMRM meetings.
- 2015 ECTRIMS Travel Grant funding the attendance at the 2015 ECTRIMS meeting in Barcelona (Spain) (EUR 400).
- 2013, 2015, 2017 Guarantors of Brain Travel Grant funding the attendance at the 2013 (GBP 800), 2015 (GBP 500) and 2017 (GBP 800) ISMRM meetings.
- 2012-2015 ISMRM Trainee Stipend supporting attendance at the annual meeting (yearly).

INVITED ORAL COMMUNICATIONS

- ISMRM Iberian Chapter 2026 *"From histology to radiological maps: a journey through MRI signal modelling and beyond"*. International Society for Magnetic Resonance in Medicine (ISMRM), 2026 annual meeting of the Iberian Chapter. **"Keynote lecture"**. University of Coimbra, Coimbra (Portugal), 02/07/2026.
- BSC 2025 *"Histology-informed computer simulations enhance Magnetic Resonance Imaging (MRI) virtual biopsies in cancer"*. **"Severo Ochoa Research Seminar"**, Barcelona Supercomputing Center (BSC), Barcelona, Spain, 13/11/2025.
- ODELIA Summer School 2025 *"An overview of Magnetic Resonance Imaging (MRI): from basic principles to cutting-edge applications"*. ODELIA consortium on open source swarm learning to empower medical AI. **"2025 ODELIA Summer School"**, Vall d'Hebron Institute of Oncology (VHIO), Barcelona, Spain, 18/09/2025.
- ISMRM Iberian Chapter 2025 *"Quantitative imaging and AI Automation"*. International Society for Magnetic Resonance in Medicine (ISMRM), 2025 annual meeting of the Iberian Chapter. **"2025 ISMRM Iberian Chapter pre-conference Summer School"**, Institut de Bioenginyeria de Catalunya (IBEC), Barcelona (Spain), 02/07/2025.
- ISMRM Workshop 2025 *"Challenges in body diffusion and how to overcome them"*. International Society for Magnetic Resonance in Medicine (ISMRM) Workshop on: **"40 years of Diffusion: past, present and future perspectives"**, Kyoto University, Kyoto (Japan), 18/02/2025.

EPFL 2024	<i>"Diffusion MR signal modelling for oncological body imaging".</i> Visitors Talk , Centre for Biomedical Imaging (CIBM), École Polytechnique Fédérale de Lausanne (EPFL), Lausanne (Switzerland), 09/10/2024.
ESMRMB workshop 2024	<i>"Modelling body microstructure".</i> ESMRMB 2024 pre-congress workshop "Leaps in Microstructure Imaging: Exploring New Horizons" , Barcelona (Spain), 02/10/2024.
DiPy workshop 2024	<i>"Advanced body diffusion MRI for oncological applications".</i> 2024 DiPy workshop , online, 14/03/2024.
ESMRMB workshop 2023	<i>"Histology-informed body diffusion MRI in oncological applications".</i> ESMRMB 2023 pre-congress workshop "Frontiers in preclinical MRI" , Basel (Switzerland), 04/10/2023.
ESMRMB lectures 2023	<i>"Diffusion MRI in the body".</i> ESMRMB Lectures on MR , "Introduction to diffusion-weighted MR imaging and spectroscopy", Cardiff (UK), 07/09/2023.
DSG ISMRM 2023	<i>"Emerging models in oncology".</i> "Virtual biopsies by diffusion MRI: are we there yet in oncology?" , Virtual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM) Diffusion Study Group (DSG). Online, 19/04/2023.
UCL 2022b	<i>"Data-driven, model-free, deep learning approach for quantitative MRI protocol design".</i> "Microstructure Imaging Meets Machine Learning" (MIML) workshop, University College London, London (UK), 13/05/2022.
ISMRM 2022	<i>"Modeling diffusion in cancer and body".</i> Educational session: "Diffusion" , 2022 annual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), London (UK), 07/05/2022.
University of Verona 2019	<i>"Diffusion MRI data harmonisation".</i> 2019 School on Brain Connectomics, University of Verona (Italy), 24/09/2019.
ISMRM Italian Chapter 2018	<i>"Axonal dispersion from diffusion MRI: a new marker of microstructural damage".</i> Italian Association for Magnetic Res. in Medicine, Padua (Italy), 10/05/2018.

MODERATION OF SCIENTIFIC SESSIONS

ECR 2026	Moderator of European Society of Radiology (ESR) Research Committee Session 10 "Real-world applications of imaging biomarkers: from hypothesis to clinical practice" , 05/03/2026, 2026 European Conference of Radiology (ECR), organised by the ESR in Vienna (Austria).
ISMRM Iberian Chapter 2025	Moderator of oral scientific session "Clinical session" , 03/07/2025, 2025 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine (ISMRM), Barcelona (Spain).
ISMRM 2022	Moderator of oral "power pitch" scientific session "Motion correction" , 10/05/2022, 2022 annual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), London (UK).
ISMRM 2021b	Moderator of oral scientific session "Microstructure: Modelling Gray & White Matter Diffusion" , 19/05/2021, 2021 virtual annual meeting of the ISMRM.
ISMRM 2021a	Facilitator of poster session "Diffusion Applications: Brain & Spine" , 18/05/2021, 2021 virtual annual meeting of the ISMRM.

PARTICIPATION IN EVALUATION PANELS

MSc thesis — UAB 2026	MSc defense at the Universitat Autònoma de Barcelona (UAB), Cerdanyola/Barcelona (Spain). Thesis title: <i>"Metabolic profiling and classification"</i>
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of brain tumours via cNMF analysis of MRS data". Exam date: 14/07/2026 (in person at UAB). Role: examiner. Degree: MSc in Bioinformatics.

Grants — Finland
2025

Grant reviewer for the Research Council of Finland, 2025 Call for Clinical Research (online evaluation).

PhD thesis —
EPFL 2024

PhD defense at the École Polytechnique Fédérale de Lausanne (EPFL), Lausanne (Switzerland). Thesis title: "*Lymph node microstructure imaging using diffusion MRI: From simulations to acquisitions*". Exam date: 08/10/2024 (in person in Lausanne). Role: external examiner. Degree: PhD. Department: Center for Biomedical Imaging (CIBM), EPFL. Thesis link [here](#).

ORGANISATION OF SCIENTIFIC EVENTS

UB Seminar 2026

Institute of Neuroscience Seminar at *Universitat de Barcelona* 2026: "[Quantifying blood-brain barrier permeability using MRI](#)", 24/04/2026. **Organisers and moderators:** Roser Sala-Llonch, Francesco Grussu.

ISMRM Iberian
Chapter 2025

ISMRM Iberian Chapter: [2025 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine \(ISMRM\)](#) (Barcelona, scheduled for 3-4/07/2025). **Organisers:** Marco Rius I (IBEC), Ortega Machuca MA (IBEC), Candiota AP (UAB), Julià-Sapé M (UAB), Muñoz Moreno E (IDIBAPS), Lope-Piedrafita S (UAB), Barba Vert I (UVic), Benito M (UCM), Grussu F (VHIO), Gómez Cabeza D (IBEC).

MIS ISMRM 2021

Member Initiated Symposium at ISMRM 2020: "[Looking Beyond Axons: Imaging the Immune System in White Matter](#)", 19/05/2021. **Organisers:** Cohen-Adad J, Grussu F, Kolind S.

WMSG ISMRM
2019

ISMRM White Matter Study Group Virtual meeting: "[Myelin Imaging in the Spinal Cord at High Field](#)", 27/06/2019, joint meeting with the High-field Study Group. **Organiser:** Grussu F. **Chair:** Cohen-Adad J.

MIS ISMRM 2019

Member Initiated Symposium at ISMRM 2019: "[Completing the Circle: Moving Multi-Parametric Neuro MRI into Clinical Practice and Trials](#)", 15/05/2019. **Organisers:** Vrenken H, Cohen-Adad J, Grussu F.

CDMRI 2018

MICCAI Workshop: Computational Diffusion MRI (CDMRI) 2018 (Granada, Spain, 20/09/2018). **Organisers:** Bonet-Carne E (UCL), Grussu F (UCL), Ning L (Harvard), Sepehrband F (USC), Tax C (Cardiff University).

MUSHAC
Challenge 2018

MICCAI Challenge: Multi-shell dMRI harmonisation and enhancement (MUSHAC, part of CDMRI 2018). **Organisers:** Bonet-Carne E (UCL), Grussu F (UCL), Ning L (Harvard), Sepehrband F (USC), Tax C (Cardiff University).

CDMRI 2017

MICCAI Workshop: Computational Diffusion MRI (CDMRI) 2017 (Quebec City, Canada, 10/09/2017). **Organisers:** Grussu F (UCL), Kaden E (UCL), Ning L (Harvard), Tax C (Cardiff University), Veraart J (NYU).

Data
Harmonisation
Challenge 2017

MICCAI Challenge: Diffusion MRI data harmonisation (part of CDMRI 2017). **Organisers:** Grussu F (UCL), Kaden E (UCL), Ning L (Harvard), Tax C (Cardiff University), Veraart J (NYU).

Spinal Cord Grey
Matter
Segmentation
Challenge 2016

Challenge: Grey Matter Segmentation: What's there and What's next? (part of SC MRI Workshop 2016). **Organisers:** Cohen-Adad J, Prados F, Landman B, Wheeler-Kingshott C, Summers P, Dupont S, Yiannakas M, Smith S, Gergely D, DeLeener B, Grussu F.

SCIENTIFIC PEER REVIEWING

2016-present

Reviewer for Bipolar Disorders; Functional Neurology; Frontiers; Human Brain Mapping; Journal of Magnetic Resonance Imaging; Journal of Neuroscience Methods; Magnetic Resonance in Medicine; NeuroImage; Medical Image

Analysis; IEEE Transactions on Medical Imaging; npj Precision Oncology; NMR in Biomedicine; Scientific Data; 2019, 2021, 2022 ISMRM annual meetings; 2021 ISMRM Iberian Chapter post-grad meeting; 2021, 2025 Iberian Chapter annual meeting; 2019, 2020 MICCAI workshop on Computational Diffusion MRI.

OFFICIAL STUDENT SUPERVISION

PhD, VHIO and UB 2023-27	Project co-supervision: <i>"Novel Magnetic Resonance Imaging biomarkers for precision medicine in oncology: integrating multi-omics and real-world data strategies"</i> . Degree: PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). Supervised in: 2023-2027 (ongoing).
PhD, VHIO and UB 2022-26	Project co-supervision: <i>"Histology-informed diffusion MRI simulations and artificial intelligence for cancer microstructure characterization"</i> . Degree: PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at the Vall d'Hebron Institute of Oncology (VHIO), Barcelona (Spain). Supervised in: 2022-2026 (ongoing).
MSc, VHIO and UAB 2025/26	Project co-supervision: <i>"Non-Invasive Tumour Microstructure Estimation Using Diffusion MRI: A Multi- Cohort Evaluation of Histo-μSim"</i> . Degree: Master's in "Bioninformatics", Universitat Autònoma de Barcelona (UAB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). Supervised in: 01/02/2026-14/07/2026 (completed). Final thesis mark: 9.69/10.
PhD, VHIO and UB 2022-26	Project co-supervision: <i>"A simulation framework for characterizing cancer microvasculature with histology-informed diffusion MRI"</i> . Degree: PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). Supervised in: 30/09/2022-23/04/2026 (completed). Final mark: excellent and <i>cum laude</i> ; international PhD award.
MEng, VHIO and TU Delft 2022/23	Project co-supervision: <i>"Unraveling tumour microstructure through diffusion MRI using histology-powered artificial intelligence"</i> . Degree: Master's in "Biomedical Engineering", Delft University of Technology (TU Delft), (Delft, The Netherlands). Project carried out at VHIO, Barcelona (Spain). Supervised in: 09-2022/06-2023 (completed).
MSc, VHIO and UI La Rioja 2022/23	Project co-supervision: <i>"Prediction of advanced biomarkers from clinical diffusion Magnetic Resonance Imaging"</i> . Degree: Master's in "Artificial Intelligence", Universidad Internacional de La Rioja (Logroño, Spain). Project carried out at VHIO, Barcelona (Spain). Supervised in: 2022/2023 (completed).
MRes, UCL 2018/19	Project co-supervision: <i>"Improving the differential diagnosis between Neuromyelitis Optica Spectrum Disorder and Multiple Sclerosis using MRI"</i> . Degree: MRes in "Clinical Neuroscience", Institute of Neurology, University College London (UCL, London, UK). Supervised in: 2018/2019 (completed).
MSc, UCL 2018/19	Project co-supervision: <i>"Evaluation of quantitative MRI indices reproducibility across scanner upgrade"</i> . Degree: MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). Supervised in: 2018/2019 (completed).
MSc, UCL 2017/18	Project co-supervision: <i>"Investigation of multi-component T₁ relaxation at 3 Tesla"</i> . Degree: MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). Supervised in: 2017/2018 (completed).
MSc, UCL 2015/16	Project co-supervision: <i>"Evaluation of strategies for co-registration between quantitative and anatomical magnetic resonance images of the human spinal cord"</i> . Degree: MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). Supervised in: 2015/2016 (completed).

TEACHING EXPERIENCE

UB Lecture	Lecture: “Advanced medical imaging: a journey through engineering, physics and computer science”. Subject/module: “Artificial intelligence in biomedical engineering”. Degree: BEng in “Biomedical engineering”, Universitat de Barcelona (Barcelona, Spain). Conveyed: 12/05/2026. Face-to-face teaching: 2 hours.
UVi-UCC Lecture and Workshop	Lecture and Workshop: “Medical image analysis”. Subject/module: “Interactomics”. Degree: MSc in “Omics Data Analysis”, Universitat de Vic-Universitat Central de Catalunya, UVic-UCC (Vic and Barcelona, Spain). Conveyed: 17/03/2026. Face-to-face teaching: 4 hours.
UCL Lecture	Lecture: “Image optimisation: SNR, CNR and sources of artifacts”. Degree: MSc in “Advanced neuroimaging”, Institute of Neurology, UCL (London, UK). Conveyed: 2019/20, 2017/18, 2016/17, 2015/16. Face-to-face teaching: 4 hours (1 hour per year). Additional tasks: exam preparation and assessment.
UCL Workshop	Workshop: Hands-on with a portable MRI scanner . Degree: MSc in “Advanced neuroimaging”, Institute of Neurology, UCL (London, UK). Conveyed: 2017/18, 2016/17, 2015/16. Face-to-face teaching: 12 hours (4 hour per year).
UniPV Workshop	Workshop: “Model fitting for quantitative MRI”. Degree: MEng in “Biomedical Engineering”, University of Pavia (Pavia, Italy). Conveyed: 2016/17. Face-to-face teaching: 2 hours. Additional tasks: exam preparation and assessment.
UCL Lecture	Lecture: “Magnetic resonance image formation”. Degree: MSc in “Advanced biomedical imaging”, Centre for Advanced Biomedical Imaging, UCL (London, UK). Conveyed: 2015/16. Face-to-face teaching: 1 hour.

OPEN SCIENCE

Code released through GitHub (<http://fragrussu.github.io> and <http://github.com/radiomicsgroup>)

GitHub 2024c	Histo-μSim: histology-informed cancer diffusion MRI (link).
GitHub 2024b	SpinFlowSim: diffusion MRI simulator in vascular networks (link).
GitHub 2024a	BodyMRITools: python code for body diffusion MRI processing (link).
GitHub 2022	MChepato: Code and synthetic data for Grussu et al, MRM 2022 (link). Available in Zenodo as record 6645258, doi: 10.5281/zenodo.6645258 .
GitHub 2020b	qMRI-Net: MRI signal model fitting based on artificial intelligence (link).
GitHub 2020a	SARDU-Net: data-driven, model-free quantitative MRI protocol design (link).
GitHub 2019b	MyRelax: tools for myelin and relaxation MRI analyses (link to the latest version). Version 1.0.0 available in Zenodo as record 4561898, doi: 10.5281/zenodo.4561898 .
GitHub 2019a	MRITools: tools for handling and managing research MRI scans (link).
GitHub 2016	StructureTensorToolbox: tools for analysis of 2D histological images (link).

Data sets released through Zenodo

Zenodo 2024b	“Histology-informed microstructural diffusion simulations for MRI cancer characterisation (Histo- μ Sim): ex vivo mouse data”. Grussu F, Grigoriou A, Macarro C and Perez-Lopez R. Data set. Zenodo record 14559355, 2024, doi: 10.5281/zenodo.14559355 .
Zenodo 2024a	“Histology-informed microstructural diffusion simulations for MRI cancer

characterisation (Histo- μ Sim): histology substrates". Grigoriou A, Macarro C, Perez-Lopez R and Grussu F. Data set. Zenodo record 14559103, 2024, doi: [10.5281/zenodo.14559103](https://doi.org/10.5281/zenodo.14559103).

DISSEMINATION TO THE WIDER PUBLIC

- 2024b Lab demonstration: "*Uso de la inteligencia artificial en la investigación contra el cáncer*" ("*Using artificial intelligence in cancer research*") at the 2024 European Researchers' Night (27/09/2024), CosmoCaixa, Barcelona (Spain).
- 2024a Video: "*Resonancia magnética para combatir el cáncer*" ("*Magnetic Resonance Imaging to fight cancer*"), released through the YouTube channel of the Vall d'Hebron Institute of Oncology (Barcelona, Spain) ([link](#), video in Spanish).
- 2022b Science fair: "*Demostración de análisis avanzado de imágenes por resonancia magnética y datos co-localizados de microscopía*" ("*Demonstration of advanced analysis of MR images and co-localised microscopy*") at the 2022 European Researchers' Night (30/09/2022), CosmoCaixa, Barcelona (Spain).
- 2022a "*Innovating Magnetic Resonance Imaging to fight diseases*". Università della Terza Età (UniTre), Mòguru, Italy (12/04/2021). Talk given to a general audience in Sardinian language.
- 2017 "*Why to get vaccinated and avoid misinformation: the scientific method in modern medicine*". Event for a general audience in Italian and Sardinian, Mòguru, Italy (17/08/2017). Organisers: Grussu F, Tur C, Coccollone E, Broccia S.
- 2015 Participation at *MS Frontiers 2015* (29-30/06/2015), organised by the UK Multiple Sclerosis Society and bringing together researchers and Multiple Sclerosis patients.
- 2013 UCL stall at *Science Uncovered*, 2013 European Researcher's Night (28/09/2013), London Science Museum.

CONFERENCE & WORKSHOP PROCEEDINGS: FIRST AUTHORSHIP

- ECR 2026 "*Response assessment and prognosis prediction in cancer immunotherapy using multi-parametric MRI: preliminary evidence*". Grussu F et al. European Conference of Radiology (ECR) 2026, C-16681 (EPOSTMoral + digital poster presentation, 07/03/2026), doi: [10.26044/ecr2026/C-16681](https://doi.org/10.26044/ecr2026/C-16681).
- ESMRMB 2024 "*Histological interpretation of Susceptibility-Perturbation MRI in human tumours of the liver*". Grussu F et al. European Society for Magnetic Resonance in Medicine and Biology (ESMRMB) 2024 (traditional poster presentation).
- ISMRM 2024b "*Histology-informed biophysical diffusion MRI model selection for enhanced liver cancer immunotherapy assessment*". Grussu F et al. International Society for Magnetic Resonance in Medicine (ISMRM) 2024 (**oral presentation**).
- ISMRM 2024a "*Two-axon population (TAP) modelling for large axon diffusion imaging in the peripheral nervous system*". Grussu F et al. International Society for Magnetic Resonance in Medicine (ISMRM) 2024 (**oral presentation**).
- ISMRM Iberian 2023 "*Extra-cellular liver diffusion modelling at high b-value: a preclinical MRI-histology study*". Grussu F et al. Iberian Chapter of the ISMRM 2023 (**oral presentation**).
- ISMRM 2022b "*Histological correlates of DR-HIGADOS microstructural metrics in the mouse and human liver*". Grussu F et al. International Society for Magnetic Resonance in Medicine (ISMRM) 2022 (power-pitch presentation).
- ISMRM 2022a "*Inter-scanner reproducibility and variability assessment of advanced liver diffusion MRI metrics*". Grussu F et al. ISMRM 2022 (d-poster presentation).

ISMRM 2021b	<i>"DR-HIGADOS: a new diffusion-relaxation framework for clinically feasible microstructural imaging of the liver"</i> . Grussu F et al. ISMRM 2021 (oral presentation, Magna cum Laude award).
ISMRM 2021a	<i>"Investigating the relationship between diffusion MRI signal cumulants and hepatocyte microstructure at fixed diffusion time"</i> . Grussu F et al. International Society for Magnetic Resonance in Medicine (ISMRM) 2021 (d-poster presentation).
ISMRM 2020	<i>"SARDU-Net: a new method for model-free, data-driven experiment design in quantitative MRI"</i> . Grussu F et al. ISMRM 2020 (power-pitch presentation, Magna cum Laude award).
ISMRM 2019	<i>"Clinically viable g-ratio imaging with unified readout at 3T: evaluation and comparison"</i> . Grussu F et al. ISMRM 2019 (e-poster presentation).
ISMRM 2018b	<i>"Magnitude versus complex-valued images for spinal cord diffusion MRI: which one is best?"</i> . Grussu F et al. ISMRM 2018 (oral presentation, Magna cum Laude award).
ISMRM 2018a	<i>"A unified signal readout improves denoising of multi-modal spinal cord MRI"</i> . Grussu F et al. ISMRM 2018 (poster presentation).
ISMRM 2017b	<i>"Origin of the time dependence of the diffusion-weighted signal in spinal cord white matter"</i> . Grussu F et al. ISMRM 2017 (oral presentation).
ISMRM 2017a	<i>"A unified signal readout for reproducible multimodal characterisation of brain microstructure"</i> . Grussu F et al. ISMRM 2017 (e-poster presentation, Magna cum Laude award, finalist at the Diffusion Study Group competition).
Brain School 2017	<i>"Whole-brain macromolecular tissue volume mapping: A comparison of imaging readouts at 3 Tesla"</i> . Grussu F et al. School of Brain Cells and Circuits "Camillo Golgi". Frontiers ISBN 978-288945-584-3 (poster presentation).
ISMRM Scientific Workshop 2016	<i>"Optimal histological quantification of neurite orientation dispersion for the validation of diffusion MRI"</i> . ISMRM Scientific workshop "Breaking the barriers of diffusion MRI" (poster presentation + Power Pitch).
ISMRM 2016	<i>"Axon diameter distribution influences diffusion-derived axonal density estimation in the human spinal cord: in silico and in vivo evidence"</i> . Grussu F et al. ISMRM 2016 (poster presentation, finalist at the Diffusion Study Group competition).
ECTRIMS 2015	<i>"Quantitative histological validation of NODDI MRI indices of neurite morphology in multiple sclerosis spinal cord"</i> . Grussu F et al. European Committee for Research and Treatment of Multiple Sclerosis (ECTRIMS) 2015 (poster presentation, short-listed for poster prize competition).
MS Frontiers 2015	<i>"Histological correlates of NODDI in the multiple sclerosis spinal cord"</i> . Grussu F et al. MS Frontiers 2015, annual scientific meeting of the UK Multiple Sclerosis Society (oral and poster presentation).
ISMRM 2015b	<i>"Quantitative histological correlates of NODDI orientation dispersion estimates in the human spinal cord"</i> . Grussu F et al. ISMRM 2015 (oral presentation, Magna cum Laude award).
ISMRM 2015a	<i>"Histological metrics confirm microstructural characteristics of NODDI indices in multiple sclerosis spinal cord"</i> . Grussu F et al. ISMRM 2015 (oral presentation, Magna cum Laude award).
British Chapter of the ISMRM 2014	<i>"Characterisation of single-shell NODDI fitting in spinal cord grey and white matter"</i> . Grussu F et al. British Chapter of the ISMRM 2014 (poster presentation).
ISMRM 2014b	<i>"Neurite orientation dispersion and density imaging of the cervical cord in vivo"</i> . Grussu F et al. ISMRM 2014 (poster presentation).
ISMRM 2014a	<i>"Single-shell diffusion MRI NODDI with in vivo cervical cord data"</i> . Grussu F et al.

ISMRM 2014 (poster presentation).

- ISMRM Workshop 2013 "In vivo estimation of neuronal orientation dispersion and density of the human spinal cord". ISMRM Scientific workshop "Multiple sclerosis as a whole-brain disease" (oral presentation).
- ISMRM 2013 "Towards spinal cord microstructure mapping with the neurite orientation dispersion and density imaging". Grussu F et al. ISMRM 2013 (poster presentation, finalist at the White Matter Study Group poster competition).

CONFERENCE & WORKSHOP PROCEEDINGS: SENIOR AUTHORSHIP

- ISMRM Iberian Chapter 2025 "Simplicity is a virtue: histology-informed model comparison selects simple diffusion representations in a colorectal cancer metastasis specimen". Grigoriou A, ..., and Grussu F. 2025 ISMRM Iberian Chapter annual meeting, p.49 (poster presentation). **2nd prize for the best poster presentation.**
- ISMRM Workshop 2025b "Which Microvascular Properties Can We Probe in Clinical Settings with Diffusion MRI?". Voronova A, ..., and Grussu F. 2025 ISMRM Workshop on 40 Years of Diffusion: Past, Present and Future Perspectives (oral presentation).
- ISMRM Workshop 2025b "Histology-Informed Microstructural Diffusion Simulations (Histo- μ Sim) for Enhanced Diffusion MRI Parameter Estimation in Cancer". Grigoriou A, ..., and Grussu F. 2025 ISMRM Workshop on 40 Years of Diffusion: Past, Present and Future Perspectives (power-pitch presentation), p. 202.
- ESMRMB 2024b "Biologically-realistic blood flow simulations reveal complex features of vascular IVIM signals". Voronova A, ..., and Grussu F. ESMRMB 2024 (traditional poster presentation).
- ESMRMB 2024a "Histology-informed cell size distribution mapping with diffusion MRI". Grigoriou A, ..., and Grussu F. ESMRMB 2024 (traditional poster presentation).
- ISMRM 2024b "FlowSim: a blood flow simulator for histology-informed diffusion MRI micro-vasculature mapping in cancer". Voronova A, ..., and Grussu F. ISMRM 2024 (power-pitch presentation, **Summa cum Laude** award, **Diffusion Study Group prize, best application (poster category)**).
- ISMRM 2024a "A Monte Carlo simulation framework for histology-informed diffusion MRI parameter estimation in cancer". Grigoriou A, ..., and Grussu F. ISMRM 2024 (power-pitch presentation, **Magna cum Laude** award).
- ISMRM Scientific Workshop 2022 "A systematic comparison of machine learning approaches for diffusion-relaxation MRI protocol enhancement in advanced solid tumours". Macarro C, ..., and Grussu F. ISMRM Workshop on Diffusion MRI From Research to Clinic 2022 (poster presentation).
- ISMRM 2019 "Sensitivity of NODDI and two-compartment SMT parameter maps in multiple sclerosis". Johnson D, ..., and Grussu F. ISMRM 2019 (e-poster presentation).

CONFERENCE & WORKSHOP PROCEEDINGS: SELECTED CO-AUTHORSHIP

- ISMRM 2023 "Decomposition of clinical ADC into intracellular and extracellular-extravascular contributions in prostate cancer using histology". Garcia-Ruiz A et al. ISMRM 2023 (oral presentation).
- ENA Symposium 2022 "Non-invasive biomarkers for response and survival prediction in patients with advanced solid tumours treated with immune checkpoint inhibitors (ICIs)". Bernatowicz K et al. European Journal of Cancer 174S1 (2022) S3–S128, doi: [10.1016/S0959-8049\(22\)00988-1](https://doi.org/10.1016/S0959-8049(22)00988-1) (EORTC-NCI-AACR (ENA) Symposium, 2022 October 26-28, Barcelona, Spain; poster presentation).

ISMRM Workshop 2022	<i>"Decoding liver intra-tumour heterogeneity with co-localized CT and multi-parametric MRI". Prior Palomares O, Grussu F, et al. ISMRM Workshop on Diffusion MRI From Research to Clinic 2022 (oral presentation).</i>
ISMRM 2022b	<i>"Deep-learning-informed parameter estimation improves reliability of spinal cord diffusion MRI". Gong T et al. ISMRM 2022 (oral presentation).</i>
ISMRM 2022a	<i>"Deep learning voxelwise classification of primary central nervous system lymphoma using DSC-PWI normalized time-intensity curves". Garcia-Ruiz A et al. ISMRM 2022 (d-poster presentation).</i>
ISMRM 2021	<i>"Associations between cervical cord sodium concentration, neuronal density and macromolecular tissue volume in spinal cord injury". Solanky B et al. ISMRM 2021 (oral presentation).</i>
ISMRM 2020	<i>"Acquiring and predicting Multi-dimensional Diffusion (MUDI) data: an open challenge". Pizzolato M et al. ISMRM 2020 (oral presentation).</i>
ISMRM 2020b	<i>"New potential MRI markers of glial scarring and tissue damage in multiple sclerosis spinal cord pathology using diffusion MRI". Palombo M et al. ISMRM 2020 (power-pitch presentation).</i>
ISMRM 2020a	<i>"SENSE reconstruction with simultaneous 2D phase correction and channel-wise noise removal (SPECTRE)". Powell E et al. ISMRM 2020 (d-poster presentation).</i>
ISMRM 2020	<i>"Quantitative MRI of the spinal cord: reproducibility and normative values across 40 sites". Alonso-Ortiz E L et al. ISMRM 2020 (oral presentation).</i>
ISMRM 2019b	<i>"Cross-scanner and cross-protocol harmonisation of multi-shell diffusion MRI data: open challenge and evaluation results". Ning L et al. ISMRM 2019 (oral presentation).</i>
ISMRM 2019a	<i>"Bound Pool Fraction mapping via steady-state MT saturation using single-shot EPI". Battiston M et al. ISMRM 2019 (oral presentation).</i>
ISMRM 2018b	<i>"Cross-vendor and cross-protocol harmonisation of diffusion MRI data: a comparative study". Tax C et al. ISMRM 2018 (oral presentation).</i>
ISMRM 2018a	<i>"Consensus acquisition protocol for quantitative MRI of the cervical spinal cord at 3T". Alley S et al. ISMRM 2018 (oral presentation).</i>
ECTRIMS 2017c	<i>"Application of Neurite Orientation Dispersion and Density Imaging (NODDI) in clinically isolated syndrome (CIS)". Collorone S et al. ECTRIMS 2017 (poster presentation).</i>
ISMRM 2017b	<i>"Boundary shift integral to compute brain and cervical spinal cord longitudinal atrophy on the same 3D T1 brain images in multiple sclerosis". Prados F et al. ISMRM 2017 (oral presentation).</i>
ISMRM 2017a	<i>"Impact of acquisition strategies and spherical deconvolution algorithms on brain connectivity mapping in early multiple sclerosis". Tur C et al. ISMRM 2017 (oral presentation).</i>
ECTRIMS 2016b	<i>"Computing spinal cord atrophy using the boundary shift integral: a more powerful outcome measure for clinical trials?". Prados F et al. ECTRIMS 2016 (poster presentation).</i>
ECTRIMS 2016a	<i>"Neurite orientation dispersion and density imaging (NODDI) reflects early microstructural brain tissue changes in clinically isolated syndrome (CIS)". Collorone S et al. ECTRIMS 2016 (poster presentation).</i>
ISMRM 2016b	<i>"Reduced field-of-view diffusion-weighted imaging of the lumbosacral enlargement: a pilot in vivo study of the healthy spinal cord using a clinical 3T MR system". Yiannakas M et al. ISMRM 2016 (e-poster presentation).</i>

ISMRM 2016a	<i>"Atrophy computation in the spinal cord using the boundary shift integral".</i> Prados F et al. ISMRM 2016 (oral presentation).
AAN 2016b	<i>"No Differences in spinal cord white and grey matter diffusion abnormalities between neuromyelitis optica spectrum disorder and multiple sclerosis".</i> Cortese R et al. American Accademy of Neurology (AAN) 2016 (poster presentation).
AAN 2016a	<i>"Neurite orientation dispersion and density imaging (NODDI) at the onset of clinically isolated syndrome (CIS): new insights in the early microstructural brain tissue changes".</i> Collorone S et al. AAN 2016 (dual presentation).
ISMRM 2015	<i>"Combined sodium-NODDI: towards quantitative in vivo intracellular and intraneurite sodium measures at 3T".</i> Solanky B et al. ISMRM 2015 (e-poster presentation).
ISMRM 2014	<i>"An investigation of brain neurite density and dispersion in multiple sclerosis using single shell diffusion imaging".</i> Magnollay L et al. ISMRM 2014 (poster presentation).
ECTRIMS 2013	<i>"Application of neurite orientation dispersion and density imaging (NODDI) to relapsing remitting multiple sclerosis (RRMS)".</i> Magnollay L et al. ECTRIMS 2013 (poster presentation).
ECTRIMS 2013	<i>"Neurite orientation dispersion and density imaging in the multiple sclerosis spinal cord".</i> Kearney H et al. ECTRIMS 2013 (e-poster presentation).
NCM 2012	<i>"Algorithms for shaping the dynamics of a bidirectional neural interface".</i> Semprini M et al. Society for the Neural Control of Movement (NCM) 2012 (poster presentation).

July 27, 2026